Emeritus Associate Professor

Thomas M. Boyd

Professors

John H. Bradford, Vice President for Global Initiatives

Brandon Dugan, Associate Department Head, Baker Hughes Chair of Petrophysics and Borehole Geophysics

Yaoguo Li

Paul C. Sava, Department Head, C.H. Green Chair of Exploration Geophysics

Jeffrey C. Shragge

Ilya D. Tsvankin

University Distinguished Professors

Kamini Singha

Ilya D. Tsvankin

Associate professors

Ge Jin

Eileen Martin

Matthew Siegfried

Assistant Professors

Bia Villas Bôas

Roslynn King

Joint appointments with loci within Geophysics

Eileen Martin, Associate Professor in Applied Mathematics and Statistics

Joint appointments with loci outside of Geophysics

Eric Anderson, Associate Professor, Civil and Environmental Engineering

Ebru Bozdag, Associate Professor, Applied Mathematics and Statistics

Elizabeth Reddy, Assistant Professor, Engineering, Design and Society

Kamini Singha, Professor, Geology and Geological Engineering

Research Associate Professors

Richard Krahenbuhl

Mengli Zhang

Affiliate Faculty

Jyoti Behura, Founder & CEO, Seismic Science LLC

Timothy Collett, Senior Scientist, US Geological Survey

Morgan Moschetti, Research Geophysicist, US Geological Survey

Ryan North, Principal Geophysicist, ISC Geoscience

Nathaniel Putzig, Senior Scientist, Planetary Science Institute

Andrei Swidinsky, Associate Professor, University of Toronto

Whitney Trainor-Guitton, Geoscience Researcher, National Renewable Energy Laboratory

David Wald, Research Geophysicist, US Geological Survey

Joint Appointments

Fred Day-Lewis, Chief Geophysicist, Pacific Northwest National Laboratory

Humanities, Arts, and Social Sciences

Degree Offered

- Master of Science in Environment, Resources & Energy Policy (non-thesis)

Certificates Offered

- Graduate Certificate in Environment, Resources & Energy Policy

Minors Offered

- Graduate Individual Minor – A 12-credit minor for graduate students pursuing degrees in other Mines academic units. Please contact either a Humanities, Arts, and Social Sciences faculty member with whom you are interested in working or the director of the ENREP program. Minor must be approved by the student's graduate committee and by the ENREP Director.

Program Description

The MS in Environment, Resources & Energy Policy (ENREP) is an interdisciplinary degree based in the Department of Humanities, Arts, and Social Sciences that trains students to be sociotechnical problem-solvers in the public policy domain. Through core courses that build a foundation in environmental, resource, and energy policy; skills courses that sharpen students' analytical and methodological toolkits; and electives from across campus that allow students to build expertise in their desired policy focus areas, students learn how to:

- Make thoughtful decisions about the future amidst social, political, environmental and technological change, as well as scientific uncertainty.
- Meaningfully engage diverse communities in policy processes and decision-making about the environment, resources, and energy.
- Analyze policy from multiple perspectives and with attention to multiple criteria, such as distributions of benefits and burdens, tradeoffs, risk, and more.
- Become leaders in the energy transition and climate change mitigation and adaptation, and recognize how these dynamics intersect other environment and resource topics.
- Communicate clearly, eloquently, and creatively in multiple professional formats and to different audiences.

Open to all undergraduate degrees and new graduates as well as early- and mid-career professionals, ENREP bridges disciplines and teaches both qualitative and quantitative skills. The program is research and

writing intensive with a strong focus on verbal and written communication. Small classes allow faculty to connect with individual interests.

ENREP prepares students for jobs in government (local, state, and federal), consulting, non-profits and non-governmental organizations, the private sector, and academia. Drawing on Mines' international reputation and on-campus networking opportunities, graduates find jobs in water management, environmental compliance, energy analytics, community outreach, sustainability strategy, energy development, climate resilience, and other areas.

This is a non-thesis MS program, requiring 30 credit-hours of coursework: 15 credit-hours in the core, and 15 hours of electives. It usually takes 1.5 years (3 semesters, including summer) to complete the program on a full-time basis.

Combined Undergraduate/Graduate Degree Programs

Mines students may earn the master's degree as part of Mines' Combined Undergraduate/Graduate Degree program. If you are interested in pursuing this route, you are encouraged to make an initial contact with the ENREP director during your sophomore year for counseling on application procedures and degree completion.

Admission Requirements

The requirements for admission are as follows:

1. An undergraduate degree with a cumulative grade-point average (GPA) at or above 3.0 (4.0 scale). Mines undergraduates need a minimum GPA of 3.0 in their Humanities, Arts and Social Sciences coursework.
2. For students whose native language is not English, Mines requires a minimum TOEFL score of 79 internet-based test (iBT) or 550 paper-based test (PBT). Tests must have been taken within the past two years to be accepted. If you have completed a university degree program in the United States or in an English-speaking country within the previous two years, you do NOT have to submit TOEFL scores.

More program and application information is available on the ENREP website: https://hass.mines.edu/natural-resources-energy-policy/.

Program Requirements

Master of Science in Environment, Resources & Energy Policy (ENREP, Non-Thesis)

The ENREP degree prepares students from multiple disciplinary backgrounds (in the social sciences and engineering) to be leaders in policy and problem-solving in the interconnected domains of environment, resources, and energy. The core curriculum engages a range of ENREP topics while offering foundational training in environment-society and resource relations, social and political change, public engagement, and policy dynamics. A variety of skills courses allow students to build further expertise in policy-relevant analytical skills to suit their desired career paths. Electives center around themes of interest to ENREP students that are also strengths of Mines faculty, including the energy transition and climate change, water management, environmental futures, international development, and more. Students engage in research- and writing-intensive assignments with a strong focus on verbal and written communication skills.

Designed for early or midcareer professionals, ENREP graduates typically find jobs in environmental, energy, or resource management/ policy at the state, federal, local or international levels; environmental or energy consulting; and energy development in various forms. Additional career paths include positions at non-profit or non-governmental organizations, government affairs, community development, academia, and more.

This is a non-thesis program, requiring 30 credit hours of coursework: 15 credit hours in the core, and 15 hours of electives. As a full-time student, it generally takes 1.5 years (three semesters) to complete the program.

Core: Foundations Courses (9 credits)

HASS578	GLOBAL ENVIRONMENTAL ISSUES	3.0
HASS584	US WATER POLITICS AND POLICY	3.0
MNGN571	ENERGY, NATURAL RESOURCES, AND SOCIETY	3.0
Total Semester Hrs		**9.0**

Core: Skills Courses (choose any two; 6 credits)

Analytics & Modeling		
EBGN590	ECONOMETRICS I	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
Community Engagement		
EDNS479	COMMUNITY-BASED RESEARCH	3.0
HASS526	INTERCULTURAL COMMUNICATION	3.0
HASS527	RISK COMMUNICATION	3.0
HASS550	POLITICAL RISK ASSESSMENT	3.0
HASS568	ENVIRONMENTAL JUSTICE	3.0
Sustainability		
CEEN501	LIFE CYCLE ASSESSMENT	3.0
CEEN595	ANALYSIS OF ENVIRONMENTAL IMPACT	3.0
HASS525	ENVIRONMENTAL COMMUNICATION	3.0
PEGN530	ENVIRONMENTAL LAW AND SUSTAINABILITY	3.0
History & Theory		
EDNS515	INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES	3.0
HASS521	ENVIRONMENTAL PHILOSOPHY	3.0

Electives (15 credits)

Pre-approved electives are organized here by theme. Students may take electives from any department, but note that some courses have prerequisites or are primarily for engineers in those fields. In those cases, students should check with the professor before taking the course. Because electives are constantly evolving, students are encouraged to review current offerings each semester. Skills courses beyond the two required also count as electives. Students may want to consider adding a minor in a related discipline that overlaps their electives.

Energy Transition

HASS590	ENERGY AND SOCIETY	3.0
HASS591	ENERGY TRANSITION: POLITICS & POLICY	3.0
EBGN502	POLITICAL ECONOMY OF THE ENERGY TRANSITION	3.0
EBGN521	MICROECONOMICS OF MINERAL AND ENERGY MARKETS	3.0

Environmental Futures

HASS521	ENVIRONMENTAL PHILOSOPHY	3.0
HASS525	ENVIRONMENTAL COMMUNICATION	3.0
HASS568	ENVIRONMENTAL JUSTICE	3.0
HASS587	ENVIRONMENTAL POLITICS AND POLICY	3.0
HASS588	GLOBAL WATER POLITICS AND POLICY	3.0
CEEN573	RECLAMATION OF DISTURBED LANDS	3.0
CEEN575	HAZARDOUS WASTE SITE REMEDIATION	3.0
CEEN597	PRACTICES AND PRINCIPLES OF ENVIRONMENTAL CONSULTING	3.0

International Development

HASS535	INTERNATIONAL DEVELOPMENT	3.0
HASS541	AFRICAN DEVELOPMENT	3.0
HASS560	GEOPOLITICS OF NATURAL RESOURCES	3.0
HASS591	ENERGY TRANSITION: POLITICS & POLICY	3.0

Minerals & Mining

CEEN556	MINING AND THE ENVIRONMENT	3.0
CEEN573	RECLAMATION OF DISTURBED LANDS	3.0
MNGN510	FUNDAMENTALS OF MINING AND MINERAL RESOURCE DEVELOPMENT	3.0
MNGN546	MINE HEALTH AND SAFETY	2.0
MNGN557	MINERAL ECONOMICS AND POLICY	2.0
MNGN562	MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY	2.0

Economics & Policy

EBGN509	MATHEMATICAL ECONOMICS	3.0
EBGN510	NATURAL RESOURCE ECONOMICS	3.0
EBGN521	MICROECONOMICS OF MINERAL AND ENERGY MARKETS	3.0
EBGN530	ECONOMICS OF INTERNATIONAL ENERGY MARKETS	3.0
EBGN535	ECONOMICS OF METAL INDUSTRIES AND MARKETS	3.0
EBGN537	ECONOMICS OF WATER	3.0
EBGN594	TIME-SERIES ECONOMETRICS	3.0

Professional Development

EBGN553	PROJECT MANAGEMENT	3.0
LICM501	PROFESSIONAL ORAL COMMUNICATION	1.0
SYGN598	SPECIAL TOPICS (PAID INTERNSHIP)	3.0
SYGN501	RESEARCH SKILLS FOR GRADUATE STUDENTS	1.0
SYGN502	INTRODUCTION TO RESEARCH ETHICS	1.0
SYGN503	TOOLS FOR SUCCESS: INTEGRATING INTO THE MINES COMMUNITY	1.0
SYGN683	ORAL COMMUNICATION SKILLS	1.0
SYGN684	WRITING SKILLS	2.0

Mines' Combined Undergraduate / Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must be taken at the 500-level, have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit. Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Program Requirements

Graduate Certificate in Environment, Resources & Energy Policy (ENREP Certificate)

Designed to be completed in a single semester, or over two semesters for part-time students, the Certificate in Environment, Resources & Energy Policy (ENREP) is a 9-credit program affiliated with the ENREP Master's program.

To earn the certificate, students must take two of the core courses for the Master's program plus an elective to be approved by the ENREP Director.

Courses

HASS521. ENVIRONMENTAL PHILOSOPHY. 3.0 Semester Hrs.
Equivalent with LAIS521,
Analyzes environmental ethics and philosophy including the relation of philosophical perspectives to policy decision making. Critically examines often unstated ethical and/or philosophical assumptions about the environment and how these may complicate and occasionally undermine productive policies. Policies that may be considered include environmental protection, economic development, and energy production and use. 3 hours seminar; 3 semester hours.
Course Learning Outcomes

HASS523. ADVANCED SCIENCE COMMUNICATION. 3.0 Semester Hrs.
Equivalent with LAIS523,
This course will examine historical and contemporary case studies in which science
communication (or miscommunication) played key roles in shaping policy outcomes and/or public perceptions. Examples of cases might include the recent controversies over hacked climate science emails, nuclear power plant siting controversies, or discussions of ethics in classic environmental cases, such as the Dioxin pollution case. Students will study, analyze, and write about science communication and policy theories related to scientific uncertainty; the role of the scientist as communicator; and media ethics. Students will also be exposed to a number of strategies for managing their encounters with the media, as well as tools for assessing their communication responsibilities and capacities. 3 hours seminar; 3 semester hours.

HASS525. ENVIRONMENTAL COMMUNICATION. 3.0 Semester Hrs.
Equivalent with LAIS525,
(I, II, S) This course explores the ways that messages about the environment and environmentalism are communicated in the mass media, fine arts, and popular culture. The course will introduce students to key readings in environmental communication, media studies, and cultural studies in order to understand the many ways in which the images, messages, and politics of environmentalism and the natural world are constructed and contested by diverse audiences. Students will critically analyze their roles as science and/or technology communicators in the context of environmental issues, and will apply their skills to creating communications projects for diverse audiences. 3 lecture hours, 3 semester hours.
Course Learning Outcomes

- Identify major events, themes, and concepts, and narratives that have shaped the modern environmental movement and societal understandings of environmentalism
- Analyze environmental debates in both academic discourse and popular culture
- Understand and engage critically with the roles that scientists and engineers play as communicators in environmental debates
- Research and develop professional written products that make strong and logical arguments using primary and secondary sources
- Sharpen oral communication and presentation skills

HASS526. INTERCULTURAL COMMUNICATION. 3.0 Semester Hrs.
The course examines intercultural communication theory and practice. In particular, the course provides students with a window into how intercultural (mis)communication cases arise, evolve, and are resolved. Students investigate communication cases and issues across a broad range of cultural divides, such as national, gender, social class, and racial/ethnic cultures. Some case studies are situated in engineering and applied science contexts.
Course Learning Outcomes

HASS527. RISK COMMUNICATION. 3.0 Semester Hrs.
How do people perceive risk, as well as make decisions and communicate under conditions of uncertainty and risk? This course explores multiple perspectives on that overarching question. Although risk perception, risk management, and risk communication are three major course components, they are not treated separately but in terms of how they interrelate. Case studies include engineers and applied scientists coping with complex forms of uncertainty and risk, communicating in organizational and public sphere contexts with multiple audiences via the press and directly to the public, stockholders, co-workers, local communities, and more. In addition, students will critically reflect on the social consequences of living with risk in our contemporary moment.
Course Learning Outcomes

HASS535. INTERNATIONAL DEVELOPMENT. 3.0 Semester Hrs.
Equivalent with LAIS535,
(I, II, S) Explores the political economy of current and recent-historical development strategies, models, efforts, and issues in various world regions. The class will focus on Africa, Asia, Eurasia, Latin America, or the Middle East, depending on the semester. Development is understood to be a nonlinear, complex set of processes involving political, economic, social, cultural, and environmental factors whose ultimate goal is to improve the quality of life for individuals. Students will explore the roles of governments, companies, organizations, and individuals. Exact topics to be covered will vary with current events and the specific region; topics might include income inequality, the role of national and private energy companies, the impact of globalization, the role of development aid, and concepts of good governance. Students may take the course up to three times, covering different regions. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- N/A

HASS541. AFRICAN DEVELOPMENT. 3.0 Semester Hrs.
Equivalent with LAIS541,
Provides a broad overview of the political economy of Africa. Its goal is to give students an understanding of the possibilities of African development and the impediments that currently block its economic growth. Despite substantial natural resources, mineral reserves, and human capital, most African countries remain mired in poverty. The struggles that have arisen on the continent have fostered thinking about the curse of natural resources where countries with oil or diamonds are beset with political instability and warfare. Readings give first an introduction to the continent followed by a focus on the specific issues that confront African development today. 3 hours lecture and discussion; 3 semester hours.

HASS550. POLITICAL RISK ASSESSMENT. 3.0 Semester Hrs.
Equivalent with LAIS550,
Uses social science analytical tools and readings as well as indices prepared by organizations, such as the World Bank and the International Monetary Fund, to create assessments of the political, social, economic, environmental and security risks that multinational corporations may face as they expand operations around the world. Students will develop detailed political risk reports for specific countries that teams collectively select. Prerequisite: HASS 545 and IPE Minor. 3 hours seminar; 3 semester hours.

HASS560. GEOPOLITICS OF NATURAL RESOURCES. 3.0 Semester Hrs.
Equivalent with LAIS560,
This seminar examines geopolitical competition between great and aspiring powers for influence, control over land and natural resources, critical geo-strategic trade routes, or even infrastructure. Using empirical evidence from case studies, students develop a deeper understanding of the interconnections between the political, economic, social, cultural and geographic dimensions of foreign policies, as well as issues of war and peace.

HASS565. SCIENCE, TECHNOLOGY, AND SOCIETY. 3.0 Semester Hrs.
Equivalent with LAIS565,
Provides an introduction to foundational concepts, themes, and questions developed within the interdisciplinary field of science and technology studies (STS). Readings address anthropological understandings of laboratory practice, sociological perspectives on the settling of techno-scientific controversies, historical insights on the development of scientific institutions, philosophical stances on the interactions between technology and humans, and relationships between science and democracy. Students complete several writing assignments, present material from readings and research, and help to facilitate discussion. 3 hours lecture and discussion; 3 semester hours.

HASS568. ENVIRONMENTAL JUSTICE. 3.0 Semester Hrs.
This course explores the history of the environmental justice movement, current and emerging environmental justice issues, and the application of environmental justice concepts and theories to environmental decision-making. Course content and activities are designed to enrich student understanding of how environmental injustice is produced (locally, regionally, and globally), how environmental justice issues are measured and analyzed, and how environmentally just outcomes can be achieved.

Course Learning Outcomes

HASS578. GLOBAL ENVIRONMENTAL ISSUES. 3.0 Semester Hrs.
Examines global issues in environment-society interactions using analytical tools and methodological approaches from the humanities, arts, and/or social sciences. Students will gain critical perspectives on global environmental problems and solutions with an emphasis on their interrelated social, political, economic, and cultural elements.
Course Learning Outcomes

- Analyze contemporary global environmental issues and how those issues intersect social, political, and/or cultural systems.
- Critically evaluate proposed solutions to global environmental issues.
- Conduct individual research and communicate findings in ways that are effective for public and policy audiences.

HASS584. US WATER POLITICS AND POLICY. 3.0 Semester Hrs.
(I) The story of water in the American West is one of engineering and applied science inextricably intertwined with a "Gordian knot" of law and policy, changing social and cultural values, and increasingly unpredictable hydrology. This course will familiarize students with the complexities of contemporary water governance, using the Colorado River system as its central case study. The Colorado River makes for an excellent point of departure because it is one of the most dammed, diverted, legislated, litigated, and loved rivers in the world and because we literally use it up; the river has seldom reached the sea since the 1960s. Indeed, the challenges that face the Colorado River's 40 million stakeholders today have less to do with applying law and engineering to developing water resources, and much more to do with figuring out how to share an over-appropriated resource while mitigating the social and ecological consequences of past choices. Our primary goal in the course will be to learn concepts of adaptive governance that provide a constructive and framework for analyzing and addressing such challenges. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Demonstrate an understanding of contemporary water politics and policy in the arid American West

HASS586. SCIENCE AND TECHNOLOGY POLICY. 3.0 Semester Hrs.
Equivalent with LAIS586,
Examines current issues relating to science and technology policy in the United States and, as appropriate, in other countries. 3 hours lecture and discussion; 3 semester hours.

HASS587. ENVIRONMENTAL POLITICS AND POLICY. 3.0 Semester Hrs.
Equivalent with LAIS587,
Explores environmental policies and the political and governmental processes that produce them. Group discussion and independent research on specific environmental issues. Primary but not exclusive focus on the U.S. 3 hours lecture and discussion; 3 semester hours.

HASS588. GLOBAL WATER POLITICS AND POLICY. 3.0 Semester Hrs.
Equivalent with LAIS588,
(II) This interdisciplinary seminar course analyzes how droughts, floods, water management, global trading system, and climate change affect the hydrological and food systems that are critically important for economic prosperity and political stability. It addresses water policy at scales that range from community level to global governance regimes. It uses relevant analytical perspectives of, for example, psychology, political economy, development studies, and institutional approaches in economic geography to help students understand how certain transboundary water conflicts have emerged, their national and regional implications, and policies and institutions that can be used to resolve them. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Understand key issues in water politics and policy in the Middle East

HASS590. ENERGY AND SOCIETY. 3.0 Semester Hrs.
Equivalent with LAIS590,
(II) The course begins with a brief introduction to global energy production and conservation, focusing on particular case studies that highlight the relationship among energy, society, and community in different contexts. The course examines energy successes and failures wherein communities, governments, and/or energy companies come together to promote socially just and economically viable forms of energy production/conservation. The course also explores conflicts driven by energy development. These case studies are supplemented by the expertise of guest speakers from industry, government, NGOs, and elsewhere. Areas of focus include questioning the forward momentum of energy production, its social and environmental impact, including how it distributes power, resources and risks across different social groups and communities. 3 hours seminar; 3 semester hours.

HASS591. ENERGY TRANSITION: POLITICS & POLICY. 3.0 Semester Hrs.
We will use social science approaches, theories, and methods to investigate the global, regional, state, and local politics of the energy transition, moving away from carbon-intensive sources to renewable and cleaner sources, including wind, solar, hydro, and nuclear power. We will look at the politics behind energy in the United States and several other countries and critically assess national and global policies to hasten the transition. Guest speakers, videos, and other course content will represent views from energy companies, non-governmental organizations, university and research entities, government representatives, and local activists. 3 lecture hours, 3 semester hours.
Course Learning Outcomes

- Compare and contrast the pros and cons of various energy sources combining energy and social science concepts such as energy density, energy security, intermittency, infrastructure costs, jobs, inter alia.
- Create an energy transition plan from fossil fuels to clean energy using energy concepts that addresses political challenges and opportunities and demonstrates knowledge of different policy instruments and how policy is made in the United States
- Compare energy production and consumption (by type of resource, amount per capita, etc.) of several countries, including at least one country in Asia and one in Africa. Assess what these differences mean for global governance of the energy transition.
- Apply social science and energy concepts to news articles from energy-focused sources.
- Differentiate between different types of research sources and when best to use each one.
- Apply professional communication best practices in written and oral forms.
- Create a research design.

HASS593. NATURAL RESOURCES & ENERGY POLICY: THEORIES AND PRACTICE. 3.0 Semester Hrs.
(I) This course introduces students to the policy-making process, drawing on a variety of theoretical approaches, geographic locations (within the US and in other countries), and resources and energy issues.

Coordinated by the NREP Graduate Director, speakers will be from HASS, Economics and Business, Petroleum Engineering, Mining, and other departments with policy expertise, as well as from others who influence and create public and private policy. In the second half of the course, students will conduct original research projects that focus on natural resources and energy, applying theoretical frameworks they have learned from the speakers. 3 lecture hours, 3 semester hours.

Course Learning Outcomes

- Identify and apply major theoretical approaches to policy
- Prepare clear and persuasive short policy briefings on natural resources and energy issues

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598A,
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598B,
.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598C,
.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598D,
.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598E,
.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598F,
.

HASS598. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS598G,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599A,
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599B,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599C,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599D,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599E,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599F,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599G,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599H,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599I,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599J,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599K,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599L,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599M,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599N,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599O,
.

HASS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS599P,
.

HASS601. ACADEMIC PUBLISHING. 2-3 Semester Hr.
Equivalent with LAIS601,
Students will finish this course with increased knowledge of general and discipline - specific writing conversations as well as the ability to use that knowledge in publishing portions of theses or dissertations. Beyond the research article, students will also have the opportunity to learn more about genres such as conference abstracts, conference presentations, literature reviews, and research funding proposals. Prerequisite: Must have completed one full year (or equivalent) of graduate school course work. Variable credit: 2 or 3 semester hours.

HASS698. SPECIAL TOPICS. 0-6 Semester Hr.
Equivalent with LAIS698A,
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

HASS699. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Equivalent with LAIS699A,
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5

to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

HASS707. GRADUATE THESIS / DISSERTATION RESEARCH CREDIT. 1-15 Semester Hr.
Equivalent with LAIS707,
(I, II, S) GRADUATE THESIS/DISSERTATION RESEARCH CREDIT Research credit hours required for completion of a Masters-level thesis or Doctoral dissertation. Research must be carried out under the direct supervision of the student's faculty advisor. Variable class and semester hours. Repeatable for credit.

LICM501. PROFESSIONAL ORAL COMMUNICATION. 1.0 Semester Hr.
A five-week course which teaches the fundamentals of effectively preparing and presenting messages. "Hands-on" course emphasizing short (5- and 10-minute) weekly presentations made in small groups to simulate professional and corporate communications. Students are encouraged to make formal presentations which relate to their academic or professional fields. Extensive instruction in the use of visuals. Presentations are rehearsed in class two days prior to the formal presentations, all of which are video-taped and carefully evaluated. 1 hour lecture/lab; 1 semester hour.

Professors

Hussein A. Amery

Lucas Bessire

Jon A. Leydens

Kenneth Osgood

Associate Professors

Tina L. Gianquitto

Kathleen J. Hancock

Adrianne Kroepsch

James D. Straker

Teaching Professors

Jonathan Cullison

Derrick Hudson, NREP Graduate Program Director

Paula A. Farca

Cortney Holles

Joseph Horan

Shannon Davies Mancus, Associate Department Head

Seth Tucker

Sandy Woodson, Department Head

Teaching Associate Professors

Masakazu Ito

Eliza Buhrer

Heather Fester

Teaching Assistant Professors

Mairead Case

Allison Kerr

Brianna Wolfe

Hennebach Visiting Assistant Professor

Angeline Letourneau

Professors Emeriti

W. John Cieslewicz

T. Graham Hereford

Carl Mitcham

Barbara M. Olds

Eul-Soo Pang

Anton G. Pegis

Thomas Philipose, University Emeritus Professor

Arthur B. Sacks

Associate Professors Emeriti

Betty J. Cannon

John Heilbrunn

Kathleen H. Ochs

Laura J. Pang

Karen B. Wiley

Teaching Professor Emeriti

James Jesudason

Robert Klimek

Teaching Associate Emerti

Rose Pass

Mechanical Engineering

PROGRAM DESCRIPTION

The graduate program in Mechanical Engineering at Mines combines a strong foundation in engineering fundamentals with applied skills to solve real-world engineering problems. The Department offers the following degrees:

- Master of Science (Non-thesis: on-campus or online modalities available)
- Master of Science (Thesis)
- Doctor of Philosophy

The Department’s course offerings and cutting-edge research are aligned with four core Mechanical Engineering focus areas: i) Biomechanics,